

PROJECT REPORT OF CO-OP PROJECT AT INDUSTRY (MODULE-2)

ON

Emotion AI: Computational Techniques for Automatic Emotion Recognition

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled **“Emotion AI - Computational Techniques for Automatic Emotion Recognition”** in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of Dr. Gurpreet Singh. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura

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Date: 16 May, 2026

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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1. Introduction

Human emotions play a crucial role in communication, decision-making, and social interactions. The ability to recognize and interpret emotions is essential for building intelligent systems that can interact naturally with humans. However, traditional computer systems lack the capability to understand emotional states, which limits their effectiveness in real-world applications.

Emotion Artificial Intelligence, also known as affective computing, is an emerging field that focuses on enabling machines to detect, analyse, and respond to human emotions. With the rapid advancement of artificial intelligence technologies, there is an increasing demand for systems that can understand user emotions in applications such as virtual assistants, healthcare monitoring, online education, and human-computer interaction systems.

Conventional emotion recognition systems primarily rely on a single modality such as facial expressions, speech signals, or textual data. However, these unimodal systems are often unreliable due to noise, ambiguity, and individual differences in emotional expression. For example, a smile may indicate happiness for one person but could represent sarcasm or discomfort for another.

To overcome these limitations, this project proposes a multimodal Emotion AI framework that integrates facial, speech, and textual data to improve accuracy and robustness. By combining multiple sources of information, the system can provide a more comprehensive understanding of human emotions.

The main objective of this project is to design and implement a scalable, accurate, and real-time emotion recognition system using deep learning techniques and adaptive fusion strategies.

2. Methodology

The proposed methodology follows a structured pipeline consisting of multiple stages to ensure accurate emotion recognition.

2.1 Data Collection

The system utilizes three different modalities:

- **Facial Data:** Collected from datasets like FER-2013 containing labeled facial expressions
- **Speech Data:** Audio signals capturing tone, pitch, and intensity
- **Text Data:** Sentences or messages representing emotional context

This multimodal approach ensures diversity and richness in input data.

2.2 Data Preprocessing

Before feeding data into models, preprocessing is performed:

- **Facial Images:**
 - Face detection
 - Image resizing
 - Normalization
- **Speech Signals:**
 - Noise removal
 - Signal normalization
 - Feature extraction using MFCC (Mel Frequency Cepstral Coefficients)
- **Text Data:**
 - Tokenization
 - Stop-word removal
 - Text embedding using NLP techniques

This step ensures clean and consistent input across all modalities.

2.3 Feature Extraction and Model Training

Each modality is processed independently using deep learning models:

- **CNN (Convolutional Neural Network):**
 - Extracts spatial features from facial expressions
 - Learns patterns like eyes, lips, and facial movements
- **LSTM (Long Short-Term Memory):**
 - Captures temporal dependencies in speech
 - Analyses variations in tone and pitch over time
- **Transformer Models:**
 - Understand contextual meaning in text
 - Capture semantic relationships in sentences

Each model outputs probability scores for emotions such as:

- Happy
- Sad

- Angry
- Neutral

2.4 Adaptive Fusion Mechanism

A key contribution of this project is the **confidence-based adaptive fusion technique**.

Instead of assigning equal importance to all modalities, the system dynamically assigns weights:

- If facial data is unclear $\rightarrow$ lower weight
- If speech is strong $\rightarrow$ higher weight

Final emotion is calculated as:

$$E_{\text{final}} = w_1 \cdot F_{\text{face}} + w_2 \cdot F_{\text{speech}} + w_3 \cdot F_{\text{text}}$$

Where:

$$w_1 + w_2 + w_3 = 1$$

This improves system reliability in uncertain conditions.

2.5 System Workflow

- Input acquisition (face, voice, text)
- Data preprocessing
- Feature extraction
- Independent model predictions
- Confidence calculation
- Adaptive fusion
- Final emotion classification

3. Tools And Technologies

The implementation of this project utilizes modern tools and frameworks:

- **Programming Language:** Python
- **Libraries:**
 - NumPy (numerical operations)

- Pandas (data handling)
- OpenCV (image processing)
- **Deep Learning Frameworks:**
 - TensorFlow
 - PyTorch
- **Models Used:**
 - CNN for image processing
 - LSTM for audio processing
 - Transformer (BERT-like) for text
- **Datasets:**
 - FER-2013 (facial expressions)
 - Speech emotion datasets
 - Text sentiment datasets

4. Implementation

The system is implemented in a modular structure:

- **Input Module:** Captures real-time data from camera, microphone, and text
- **Preprocessing Module:** Cleans and formats data
- **Model Module:** Processes each modality independently
- **Fusion Module:** Combines outputs using adaptive weighting
- **Output Module:** Displays predicted emotion

The system is capable of real-time processing with low latency, making it suitable for applications like:

- Virtual assistants

- Smart learning systems
- Healthcare monitoring

5. Major Results/ Findings

The experimental results demonstrate that the proposed multimodal Emotion AI framework significantly outperforms traditional unimodal approaches. The unimodal models achieve an average accuracy in the range of **70–80%**, whereas the proposed multimodal system achieves over 90% accuracy, resulting in an improvement of approximately 15–20%. This improvement is achieved by combining facial, speech, and textual data, which provide complementary emotional information.

A key finding of this study is the effectiveness of the confidence-based adaptive fusion mechanism, which dynamically assigns weights to each modality based on its reliability. This enables the system to maintain stable performance even in noisy or uncertain conditions, unlike unimodal systems whose accuracy drops significantly.

The system performs well across different emotional classes, with higher accuracy for *happy* and *neutral* emotions, while minor misclassification is observed between similar emotions such as sad and fear. Furthermore, the model shows consistent performance across evaluation metrics such as accuracy, precision, recall, and F1-score, along with acceptable latency for real-time applications. These findings validate the effectiveness and robustness of the proposed multimodal framework over traditional methods.

6. Conclusion

The experimental study concludes that the proposed multimodal Emotion AI framework provides a significant improvement in emotion recognition performance by integrating facial, speech, and textual data. The use of deep learning models such as CNN, LSTM, and Transformer enables effective feature extraction from different modalities, while the confidence-based adaptive fusion mechanism enhances overall accuracy and robustness.

The system successfully overcomes the limitations of unimodal approaches by maintaining stable performance even in noisy and uncertain conditions. With an accuracy improvement of approximately **15–20%**, the framework demonstrates its effectiveness for real-time emotion

detection and practical applications such as healthcare monitoring, e-learning systems, and intelligent virtual assistants.

7. Future Scope

Future work can focus on improving the computational efficiency of the system to reduce processing time and resource requirements. The use of larger and more diverse datasets can further enhance model generalization and accuracy. Additionally, advanced fusion techniques and real-time optimization methods can be explored to improve performance.

Further research can also address important challenges related to data privacy, security, and ethical concerns in emotion recognition systems. The proposed framework can be extended and deployed in real-world applications such as mental health monitoring, human-computer interaction, and smart environments.

8. References

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